

Mathematics Bootcamp

Introduction to Proofs and Choice Theory

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Today

- Formal Language and Proof Structure
- Logical Connectives and Quantifiers
- Sets and Binary Relations
- Preference-Based and Choice-Based Decision Models
 - Rationality, WARP, and Rationalization

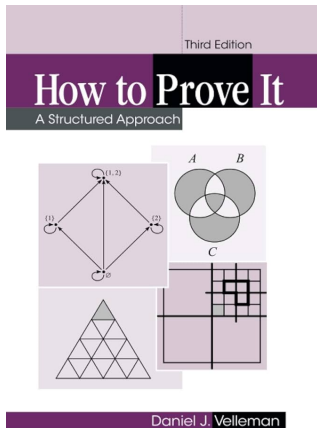
Logic and Precision

- From Ordinary Language to Scientific Language.
- Symbols, Ideas and the Real world.
- Ideas, Truth Value and Obscurantism.

These Slides

- In general, I will use the slides to save time and keep notation for the theorems: straight book screenshots.
- But most of the work will be in the board.
We learn a language by practice: trial and error.
- Proofs are done on the board; they are collected in a separate handout.
- Some believe that studying measure theory is a great way to start. This is like syntactic analysis for a baby in his first words.

Just Prove It.



- Mathematical statements have a **truth value**: each is either true or false.
- Logic tells us which conclusions follow from stated assumptions.
- Most of what follows is taken from Velleman's book.

Structured Programming

do

if [condition]

[List of instructions goes here.]

else

[Alternative list of instructions goes here.]

while [condition]

Proofs as Structured Programming

Let x be arbitrary.

Suppose $P(x)$ is true.

[Proof of $Q(x)$ goes here.]

Thus, if $P(x)$ then $Q(x)$.

Thus, for all x , if $P(x)$ then $Q(x)$.

Logical Connectives

Conjunction: $P \wedge Q$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction: $P \vee Q$

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Negation: $\neg P$

P	$\neg P$
T	F
F	T

Quantifiers and Negation

Quantifiers

Fix a domain X . A quantifier binds the variable x :

$\forall x \in X P(x) :$ $P(x)$ holds for every $x \in X$,

$\exists x \in X P(x) :$ $P(x)$ holds for at least one $x \in X$.

An occurrence of x outside the scope of a quantifier is *free*.

Negation laws ($\iff$ means “logically equivalent”)

$$\neg(P \wedge Q) \iff \neg P \vee \neg Q,$$

$$\neg(P \vee Q) \iff \neg P \wedge \neg Q,$$

$$\neg[\exists x \in X P(x)] \iff \forall x \in X \neg P(x), \quad \neg[\forall x \in X P(x)] \iff \exists x \in X \neg P(x).$$

1. Implication: $P \rightarrow Q$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

2. The Contrapositive

P	Q	$\neg Q$	$\neg P$	$\neg Q \rightarrow \neg P$	$P \rightarrow Q$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

3. Proof by Contradiction

P	Q	$\neg Q$	$P \wedge \neg Q$	$\neg(P \wedge \neg Q)$	$P \rightarrow Q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	F	T	T
F	F	T	F	T	T

Method for proving $P \rightarrow Q$

Assume P and $\neg Q$. If these assumptions imply a contradiction, written $\rightarrow\leftarrow$, then $P \wedge \neg Q$ is impossible. Therefore $P \rightarrow Q$.

4. Mathematical Induction

Bartle 1.2.3 Principle of Mathematical Induction

Let $\mathbb{N} = \{1, 2, 3, \dots\}$ and fix $n_0 \in \mathbb{N}$. For each natural number $n \geq n_0$, let $P(n)$ be a statement. Suppose that:

1. $P(n_0)$ is true.
2. For every $k \geq n_0$, if $P(k)$ is true, then $P(k + 1)$ is true.

Then $P(n)$ is true for every natural number $n \geq n_0$.

From Logic to Set Theory

Definition 1.4.1

The *intersection* of two sets A and B is the set $A \cap B$ defined as follows:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

The *union* of A and B is the set $A \cup B$ defined as follows:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

The *difference* of A and B is the set $A \setminus B$ defined as follows:

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

Set Theory

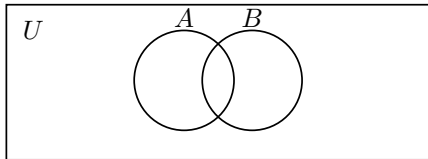


Figure 1.7.

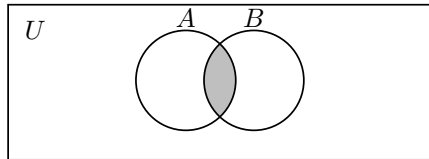


Figure 1.8. $A \cap B$.

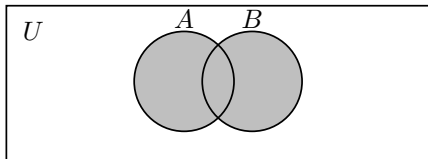


Figure 1.9. $A \cup B$.

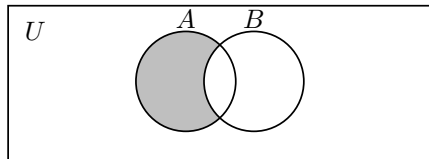


Figure 1.10. $A \setminus B$.

Set Theory

Definition 1.4.5

Suppose A and B are sets. A is a *subset* of B , written $A \subseteq B$, if every element of A is also an element of B . A and B are *disjoint* if they have no elements in common, that is, if $A \cap B = \emptyset$.

- Given $A \subseteq U$, its *complement in U* is

$$A^c = U \setminus A = \{x \in U \mid x \notin A\}.$$

- De Morgan's Laws also apply in Sets:

$$(A \cup B)^c = A^c \cap B^c,$$

$$(A \cap B)^c = A^c \cup B^c.$$

Set Theory

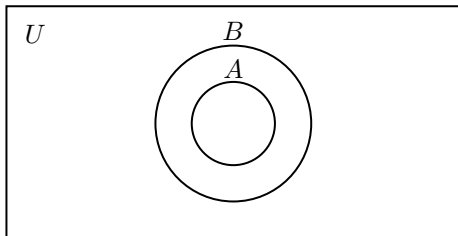


Figure 1.14. $A \subseteq B$.

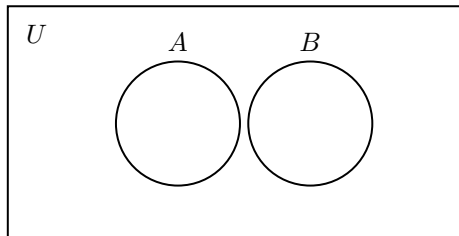


Figure 1.15. $A \cap B = \emptyset$.

Set Theory

Definition 4.1.1

For sets A and B , the *Cartesian product* $A \times B$ is the set of ordered pairs whose first coordinate lies in A and second in B :

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Binary Relations

Definition 4.2.1

Suppose A and B are sets. A set $R \subseteq A \times B$ is a *relation from A to B* . A *relation on A* is a set $R \subseteq A \times A$.

Properties of a Relation on A

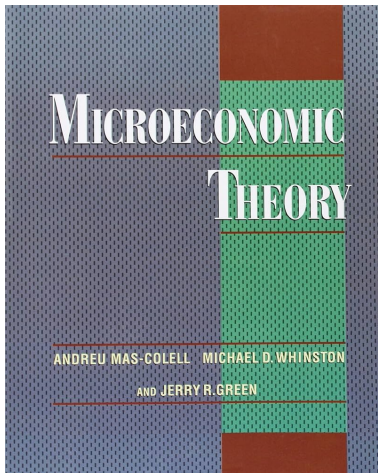
For a relation R on A :

1. *Reflexive*: xRx for every $x \in A$.
2. *Complete*: xRy or yRx (or both) for every $x, y \in A$.
3. *Symmetric*: $xRy \rightarrow yRx$ for every $x, y \in A$.
4. *Transitive*: $(xRy \wedge yRz) \rightarrow xRz$ for every $x, y, z \in A$.

Time for Economics

- We have seen some mathematical objects and their properties.
- More importantly, these are the words in the formal language that we use in economics.
- Let's try to read some economics.

MWG: Microeconomic Theory



- Direct Screenshots:
Feel that you can learn from it.
- This should not be your unique resource.
- In my 1st year, I also used:
 - **Levin Teaching Notes**
 - Rubinstein Notes Book
 - Kreps Books: 1990, 2012

Individual Decision Making

C H A P T E R

1

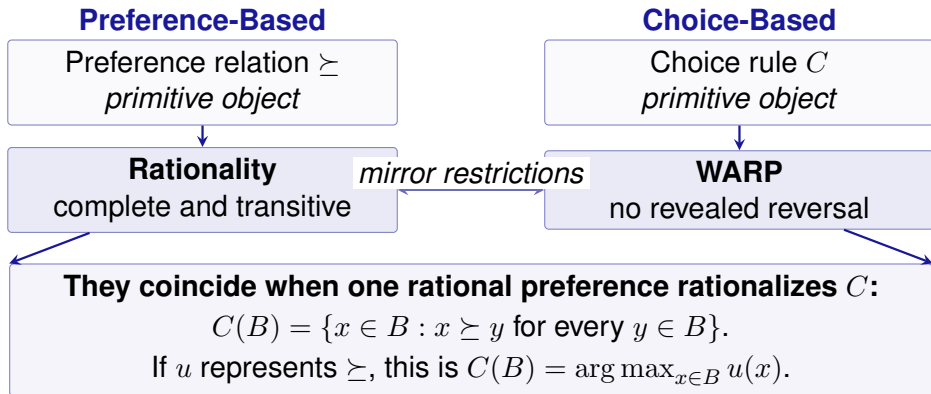
Preference and Choice

1.A Introduction

In this chapter, we begin our study of the theory of individual decision making by considering it in a completely abstract setting. The remaining chapters in Part I develop the analysis in the context of explicitly economic decisions.

The starting point for any individual decision problem is a *set of possible (mutually exclusive) alternatives* from which the individual must choose. In the discussion that follows, we denote this set of alternatives abstractly by X .

The Two Approaches



Preference as the Primitive

A Preference Relation Is the Starting Point

A preference relation $\succeq$ on X records the decision maker's ranking:

$$x \succeq y \iff x \text{ is at least as good as } y.$$

- **Rationality** restricts it: complete and transitive.
- A **utility function** represents it: $x \succeq y \iff u(x) \geq u(y)$.

Preference-Based Approach

1.B Preference Relations

In the preference-based approach, the objectives of the decision maker are summarized in a *preference relation*, which we denote by $\succsim$. Technically, $\succsim$ is a binary relation on the set of alternatives X , allowing the comparison of pairs of alternatives $x, y \in X$. We read $x \succsim y$ as “ x is at least as good as y .” From $\succsim$, we can derive two other important relations on X :

- (i) The *strict preference* relation, $\succ$, defined by

$$x \succ y \Leftrightarrow x \succsim y \text{ but not } y \succsim x$$

and read “ x is preferred to y .”¹

- (ii) The *indifference* relation, $\sim$, defined by

$$x \sim y \Leftrightarrow x \succsim y \text{ and } y \succsim x$$

and read “ x is indifferent to y .”

Preference-Based Approach

In much of microeconomic theory, individual preferences are assumed to be *rational*. The hypothesis of rationality is embodied in two basic assumptions about the preference relation $\succsim$: *completeness* and *transitivity*.²

Definition 1.B.1: The preference relation $\succsim$ is *rational* if it possesses the following two properties:

- (i) *Completeness*: for all $x, y \in X$, we have that $x \succsim y$ or $y \succsim x$ (or both).
- (ii) *Transitivity*: For all $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

- *Do not waste any attention* speculating about rationality (yet!).
- **Assume** a DM with rational preferences and see where it leads...

This is exactly the approach we follow.

Preference-Based Approach

Proposition 1.B.1: If $\succsim$ is rational then:

- (i) $\succ$ is both *irreflexive* ($x \succ x$ never holds) and *transitive* (if $x \succ y$ and $y \succ z$, then $x \succ z$).
- (ii) $\sim$ is *reflexive* ($x \sim x$ for all x), *transitive* (if $x \sim y$ and $y \sim z$, then $x \sim z$), and *symmetric* (if $x \sim y$, then $y \sim x$).
- (iii) if $x \succ y \succsim z$, then $x \succ z$.

- We can prove this proposition. Let's do it.

Choice as the Primitive

A Choice Rule Is the Starting Point

Let $\mathbb{B}$ be the domain of observed menus. A choice rule satisfies

$$C : \mathbb{B} \rightarrow 2^X \setminus \{\emptyset\}, \quad \emptyset \neq C(B) \subseteq B.$$

Choice-Based Approach

1.C Choice Rules

In the second approach to the theory of decision making, choice behavior itself is taken to be the primitive object of the theory. Formally, choice behavior is represented by means of a *choice structure*. A choice structure $(\mathcal{B}, C(\cdot))$ consists of two ingredients:

(i) $\mathcal{B}$ is a family (a set) of nonempty subsets of X ; that is, every element of $\mathcal{B}$ is a set $B \subset X$. By analogy with the consumer theory to be developed in Chapters 2 and 3, we call the elements $B \in \mathcal{B}$ *budget sets*.

(ii) $C(\cdot)$ is a *choice rule* (technically, it is a correspondence) that assigns a nonempty set of chosen elements $C(B) \subset B$ for every budget set $B \in \mathcal{B}$. When $C(B)$ contains a single element, that element is the individual's choice from among the alternatives in B . The set $C(B)$ may, however, contain more than one element. When it does, the elements of $C(B)$ are the alternatives in B that the decision maker *might* choose; that is, they are her *acceptable alternatives* in B . In this case, the set $C(B)$

WARP and Revealed Preference

Definition 1.C.1: Weak Axiom of Revealed Preference

If $x, y \in B$ and $x \in C(B)$, then for every $B' \in \mathbb{B}$ with $x, y \in B'$,

$$y \in C(B') \implies x \in C(B').$$

Definition 1.C.2: Revealed Preference

$$x \succeq^* y \iff x, y \in B \text{ and } x \in C(B) \text{ for some } B \in \mathbb{B}.$$

Read: x is *revealed at least as good as* y . Informally, x is *revealed preferred* to y if in addition $y \notin C(B)$.

A Rich Domain of Menus

Every Small Menu Is Observable

The domain is *rich* if every nonempty $B \subseteq X$ with $|B| \leq 3$ belongs to $\mathbb{B}$. Thus, for distinct $x, y, z \in X$, it contains

$$\{x\}, \{y\}, \{z\}; \quad \{x, y\}, \{x, z\}, \{y, z\}; \quad \{x, y, z\}.$$

- **Pairs** recover pairwise comparisons.
- **Triples** test them for transitivity.

Rationalizing Choice

Rationalization

A preference relation $\succeq$ *rationalizes* a choice rule C on $\mathbb{B}$ if, for every $B \in \mathbb{B}$,

$$C(B) = \{x \in B : x \succeq y \text{ for every } y \in B\}.$$

The choice rule is *rationalizable* if some rational preference relation rationalizes it.

WARP and Rationalizability

Theorem

Suppose $\mathbb{B}$ contains every nonempty subset of X with at most three alternatives, and $C(B) \neq \emptyset$ for every $B \in \mathbb{B}$. Then C is rationalizable if and only if it satisfies WARP.

- **Pairs** make $\succ^*$ complete.
- **Triples** make $\succ^*$ transitive.

Demand on Walrasian Menus

Restrict the Menu Domain

Consumer demand evaluates the same choice rule on Walrasian menus:

$$B(p, I) = \{x \in \mathbb{R}_+^L : p \cdot x \leq I\}, \quad x(p, I) := C(B(p, I)).$$

When C is rationalized by $\succeq$ and represented by u ,

$$x(p, I) = C_{\succeq}(B(p, I)) = \arg \max_{x \in B(p, I)} u(x).$$

Important. The converse above used a rich domain. Walrasian budgets are restricted, so WARP alone does not reproduce it.

Day 2 supplies the analysis that makes $\arg \max$ well defined.